

Jialong Wu

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EDUCATION

Ph.D. in Software Engineering

Aug 2022 – June 2027 (*expected*)

School of Software, Tsinghua University

Beijing, China

- Advisor: [Prof. Mingsheng Long](#)
- Research Interests: World Models, Reinforcement Learning

Bachelor in Software Engineering

Aug 2018 – June 2022

School of Software, Tsinghua University

Beijing, China

- GPA: 3.98/4.0, Rank: 1/89

Bachelor in Pure and Applied Mathematics (*Second Degree*)

Aug 2019 – June 2022

Department of Mathematical Sciences, Tsinghua University

Beijing, China

- GPA: 3.75/4.0

EXPERIMENT

Research Intern

March 2025 – *present*

ByteDance Seed

Beijing, China

PUBLICATIONS AND PREPRINTS

RLVR-World: Training World Models with Reinforcement Learning

Jialong Wu, Shaofeng Yin, Ningya Feng, Mingsheng Long#. *arXiv preprint*, 2025.

[[Project website](#)] [[PDF](#)] [[Code](#)] [[Models & Datasets](#)]

Vid2World: Crafting Video Diffusion Models to Interactive World Models

Siqiao Huang*, Jialong Wu*, Qixing Zhou, Shangchen Miao, Mingsheng Long#. *arXiv preprint*, 2025.

[[Project website](#)] [[PDF](#)]

Seed1.5-VL Technical Report

Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, ... , Jialong Wu (*Contributor*), et al. *arXiv preprint*, 2025.

[[Project website](#)] [[PDF](#)] [[Code](#)]

Trajectory World Models for Heterogeneous Environments

Shaofeng Yin*, Jialong Wu*, Siqiao Huang, Xingjian Su, Xu He, Jianye Hao, Mingsheng Long#. *ICML*, 2025.

[[PDF](#)] [[Code](#)]

CompilerDream: Learning a Compiler World Model for General Code Optimization

Chaoyi Deng*, Jialong Wu*, Ningya Feng, Jianmin Wang, Mingsheng Long#. *KDD*, 2025.

[[PDF](#)] [[Code](#)]

Long-Sequence Recommendation Models Need Decoupled Embeddings

Ningya Feng*, Junwei Pan*, Jialong Wu*, Baixu Chen, Ximei Wang,

Qian Li, Xian Hu, Jie Jiang, Mingsheng Long#. *ICLR*, 2025.

[[PDF](#)] [[Code](#)]

iVideoGPT: Interactive VideoGPTs are Scalable World Models

Jialong Wu*, Shaofeng Yin*, Ningya Feng, Xu He, Dong Li, Jianye Hao, Mingsheng Long#. *NeurIPS*, 2024.

[[Project Website](#)] [[PDF](#)] [[Code](#)] [[Models](#)] [[Poster](#)] [[Slides](#)] [[Blog](#)]

HarmonyDream: Task Harmonization Inside World Models

Haoyu Ma*, Jialong Wu*, Ningya Feng, Chenjun Xiao, Dong Li, Jianye Hao, Jianmin Wang, Mingsheng Long#. *ICML*, 2024.

[[PDF](#)] [[Code](#)] [[Poster](#)] [[Slides](#)] [[Talk](#)] [[Blog](#)]

Pre-training Contextualized World Models with In-the-wild Videos for Reinforcement Learning

Jialong Wu*, Haoyu Ma*, Chaoyi Deng, Mingsheng Long#. *NeurIPS*, 2023.

[[PDF](#)] [[Code](#)] [[Poster](#)] [[Slides](#)] [[Blog](#)]

CLIPood: Generalizing CLIP to Out-of-Distributions

Yang Shu*, Xingzhuo Guo*, **Jialong Wu**, Ximei Wang, Jianmin Wang, Mingsheng Long#. *ICML*, 2023.
[\[PDF\]](#) [\[Code\]](#)

Supported Policy Optimization for Offline Reinforcement Learning

Jialong Wu, Haixu Wu, Zihan Qiu, Jianmin Wang, Mingsheng Long#. *NeurIPS*, 2022.
[\[PDF\]](#) [\[Code\]](#) [\[Poster\]](#) [\[Slides\]](#)

Out-of-Dynamics Imitation Learning from Multimodal Demonstrations

Yiwen Qiu, **Jialong Wu**, Zhangjie Cao, Mingsheng Long#. *CoRL*, 2022.
[\[PDF\]](#) [\[Code\]](#)

Flowformer: Linearizing Transformers with Conservation Flows

Haixu Wu, **Jialong Wu**, Jiehui Xu, Jianmin Wang, Mingsheng Long#. *ICML*, 2022.
[\[PDF\]](#) [\[Code\]](#)

HONORS AND AWARDS

ICLR 2025 Notable Reviewer	2025
ICLR 2025 [Link]	
National Scholarship (Top 2%)	2020, 2024
Ministry of Education of the People's Republic of China [Certificate 2020] [Certificate 2024]	
Excellent Teaching Assistant Award	2024
Tsinghua University [Certificate]	
NeurIPS 2023 Top Reviewer	2023
NeurIPS 2023 [Link]	
Future Researcher Scholarship	2022
Tsinghua University [Certificate]	
Outstanding Graduates (Top 2%)	2022
School of Software [Certificate] & Tsinghua University [Certificate] & Beijing [Certificate]	
Excellent Graduation Thesis	2022
Tsinghua University [Certificate]	
Comprehensive Excellence Award (Top 10%)	2019, 2021, 2023
Tsinghua University [Certificate 2019] [Certificate 2021] [Certificate 2023]	
34th National Olympiad in Informatics (NOI 2017), Silver Medal (81st Place)	July 2017
China Computer Federation (CCF) [Certificate] [Link]	

INVITED TALKS AND ACTIVITIES

A Path Towards General World Models	June, 2025
BAAI Conference , Beijing, China	
Panelist	Feb, 2025
AIFC Side Event of the AI Action Summit 2025 , Paris, France	
Towards General World Models: Pre-training, Multi-Modality, and Scalable Architecture	June, 2024
ByteDance Research	
Intrinsic Task Harmonization Inside World Models	Feb, 2024
RLChina	

PROFESSIONAL SERVICES

Reviewer | PC Member

- International Conference on Learning Representations (*ICLR*) 2024 – 2025
- Conference on Neural Information Processing Systems (*NeurIPS*) 2023 – 2025
- International Conference on Machine Learning (*ICML*) 2022 – 2025
- Transactions on Machine Learning Research (*TMLR*)

Teaching Assistant

- Deep Learning, Fall 2023, Tsinghua University. Instructor: Prof. Mingsheng Long.
- Introduction to Algorithms, Spring 2023, Tsinghua University. Instructor: Prof. Bin Wang.
- Machine Learning, Fall 2022, Tsinghua University. Instructor: Prof. Mingsheng Long.
- Introduction to Artificial Intelligence, Spring 2021, 2022, 2024, Tsinghua University. Instructor: Prof. Mingsheng Long.

Student Counselor

- Undergraduate Student Counselor, Dushi College, Tsinghua University, 2024–2027 (*expected*).

SELECTED PROJECTS

Awesome Multi-task Learning | 760+ Stars

Up-to-date

A curated list of datasets, codebases, and papers on Multi-Task Learning (MTL).

ImageLabeler | C++, Qt | 120+ Stars, 40+ Forks

Summer 2019

Image annotation tool, supporting 2D/3D detection/segmentation annotation.

Incremental Decision Tree | Python (*NumPy*, *Matplotlib*, *Pandas*)

Fall 2020

Re-implementation of VFDT and EFDT, reproducing paper *Extremely Fast Decision Tree*.

TECHNICAL SKILLS

Programming Language: Python, C/C++, MATLAB, Java, HTML/CSS, JavaScript

Professional Software: PyTorch, JAX, NumPy, Git, LaTeX, Vue

Language: China (native), English

Last updated: August 28, 2025